AMAN KUMAR SINGH

B. Tech in Computer Science Engineering

2023-2027

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Github

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Portfolio

PROFESSIONAL SUMMARY

Full Stack Developer skilled in MERN stack and cloud technologies, with experience building scalable web apps and RESTful APIs. Strong in both frontend (React, Next.js, Tailwind CSS) and backend (Node.js, MongoDB) development. Passionate about automation, AI integration, and solving real-world problems through efficient code. Delivered many projects, including AIRBNB Replica and AI Cyber Threat Detection. Quick learner with strong problem-solving skills and a commitment to clean, maintainable code.

EDUCATION

Vellore Institute of Technology University, Chennai *B. Tech in Computer Science Engineering*	8.67	Aug'23 - May'27 Chennai, TN
Vig English School, Jamshedpur *Senior Secondary School, ISC*	91.50%	June 2022 Jamshedpur, Jharkhand
Vig English School, Jamshedpur *High School, ICSE*	92.60%	May 2020 Jamshedpur, Jharkhand

PROJECTS

- **Infrared Proximity Sensor**
  - Demonstrated real-world usability by simulating a **parking assistance system** or **touchless object counter**.
  - Used an **IR transmitter-receiver pair** (IR LED & photodiode) to emit and detect reflected infrared signals.
  - Integrated with an **Arduino microcontroller** to process sensor data and trigger responses (e.g., LED alerts or servo motor control).

- **Airbnb Replica** - Repository
  - **MERN** Stack Travel Platform: Airbnb-like web app with user auth, booking, and admin dash
  - Smart Property Search: Filters, Google Maps, image galleries, and pagination
  - Secure Payments: Stripe integration with booking dates and email confirmations
  - Host/Admin Tools: **CRUD** operations for listings and user management
  - Optimized Performance: **Cloudinary** image handling + responsive Tailwind design.

- **Network anomaly detection** - Repository
  - It analyzes network traffic, detects anomalies, and provides visualization tools.
  - It is a group project built using machine learning **(Isolation Forest).**
  - Built with **Python**, **Flask**, **Scikit-learn**, and uses **Wireshark/tshark** for data capture.

- **TripOnTheGo** – Repository
  - A hotel and travel booking platform that enables users to search for hotels by city, view listings, and make reservations.
  - Includes a feature-rich **admin dashboard** for managing hotel listings, bookings, and customer data.
  - Built using **PHP, MySQL, HTML, CSS, and JavaScript**, and runs on a local server via **XAMPP**.
  - Offers seamless customer interaction with a user login system and real-time data handling through a connected database.

SKILLS

SOFT SKILLS: Critical Thinking, Team Collaboration, Communication, Leadership, Time Management, Adaptability, Problem-Solving, Project Management, Creativity, Attention to Detail
Programming Languages: C++, C, JAVA, Python, JavaScript
Frameworks & Libraries: HTML, CSS, React.js, NodeJS, MongoDB
Development Tools/IDEs: VS Code, IntelliJ IDEA, PyCharm, Git/GitHub
Version Control: Git, GitHub, GitLab
Databases: SQL, MySQL, NoSQL, MongoDB
APIs & Web Services: RESTful APIs, Express.js

CERTIFICATIONS

- Full-Stack Web Development (MERN) - Certificate
- Cyber Security Basics by ISACA Chennai Chapter - Certificate
- Spoken Tutorials Training in C, CPP, Python by IIT BOMBAY - C , C++ , Py
- TCS iON Career Edge - Young Professional - Certificate